

2012
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 13, 17, 19, 23, 25 and 27

Total marks – 100

Section I Pages 2–28

75 marks

This section has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this part

Part B – 55 marks

- Attempt Questions 21–30
- Allow about 1 hour and 40 minutes for this part

Section II Pages 29–39

25 marks

- Attempt ONE question from Questions 31–35
- Allow about 45 minutes for this section

Section I

75 marks

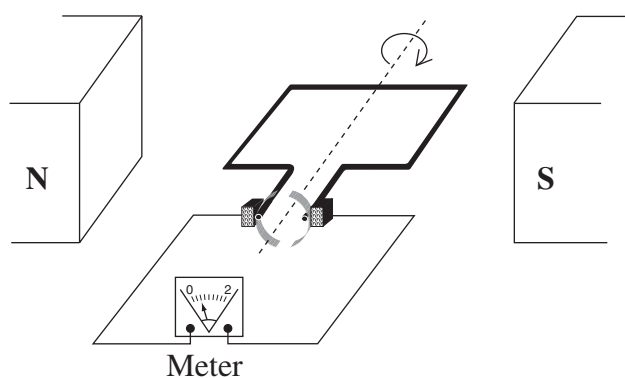
Part A – 20 marks

Attempt Questions 1–20

Allow about 35 minutes for this part

Use the multiple-choice answer sheet for Questions 1–20.

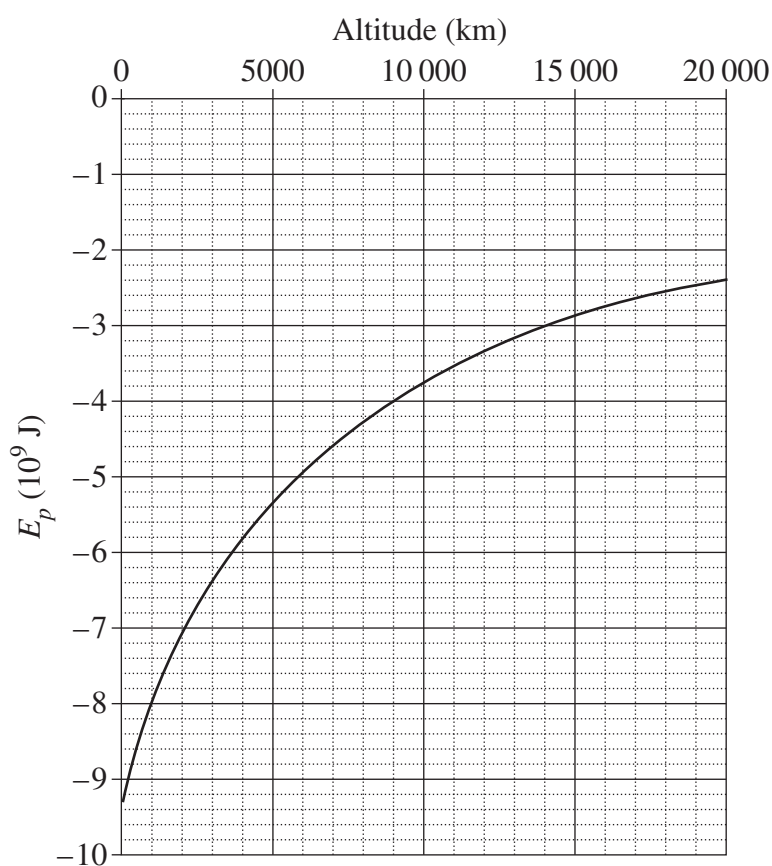
- 1 The diagram shows a device connected to a meter.



What device is shown in the diagram?

- (A) AC motor
 - (B) DC motor
 - (C) AC generator
 - (D) DC generator
- 2 What is currently used to define the standard metre?
- (A) The speed of light
 - (B) The signals from GPS satellites
 - (C) The wavelength of light from a krypton lamp
 - (D) The distance between two lines on a platinum iridium bar

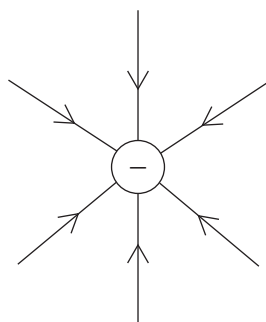
- 3 What part of a cathode ray tube allows a person to observe the position of an electron beam?
- (A) A maltese cross
(B) A heated filament
(C) A fluorescent screen
(D) A uniform magnetic field
- 4 The graph shows how the gravitational potential energy (E_p) of a satellite changes with its altitude.



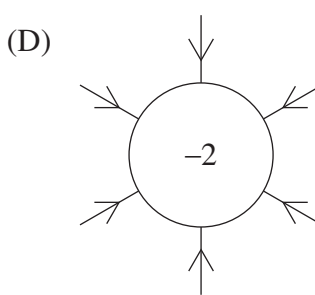
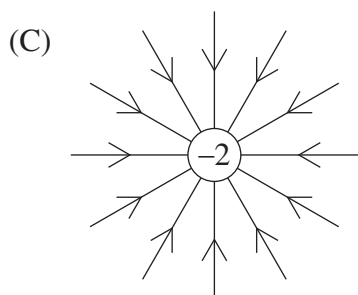
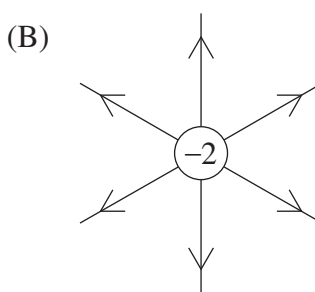
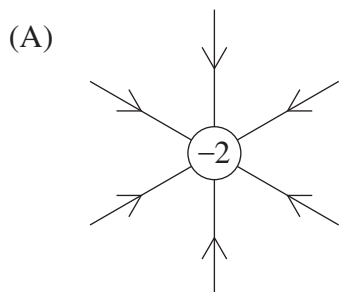
What is the change in gravitational potential energy of the satellite when its altitude is reduced from 14 000 km to 4000 km?

- (A) -8.8×10^9 J
(B) -2.8×10^9 J
(C) 2.8×10^9 J
(D) 8.8×10^9 J

- 5 Which of the following could be added to a crystal of silicon to create a p-type semiconductor?
- (A) Carbon
 - (B) Copper
 - (C) Gallium
 - (D) Phosphorus
- 6 The diagram represents the electric field around a negative charge.



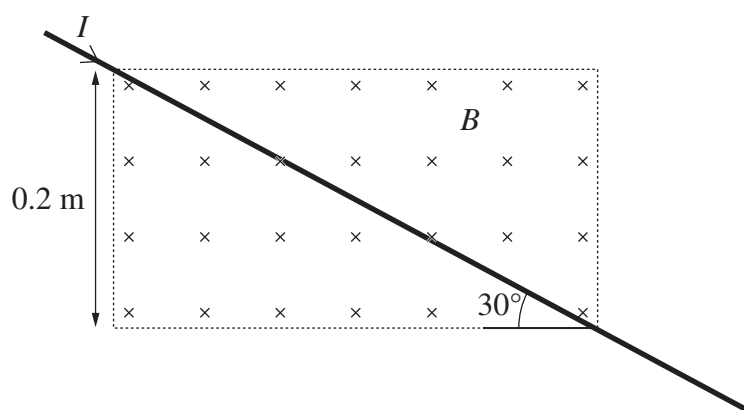
If the magnitude of the charge were doubled, which diagram would best represent the new electric field?



- 7 In moving-coil galvanometers, the coil experiences a torque because of the current flowing through it.

What ensures that this torque is directly proportional to the current flowing through the coil?

- (A) A return spring
(B) A radial magnetic field
(C) A laminated soft iron core
(D) A constant cross-sectional area of the coil
- 8 A current-carrying wire passes through a region of uniform magnetic field, magnitude 0.05 T, and as a result experiences a force of magnitude 0.03 N.

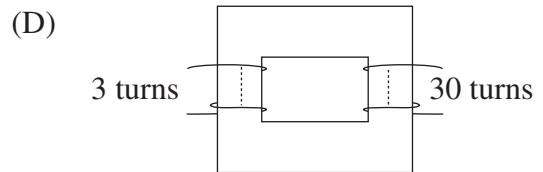
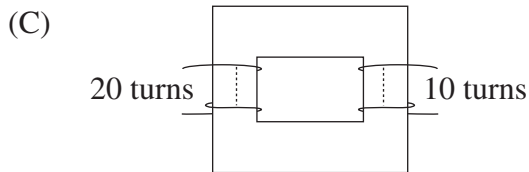
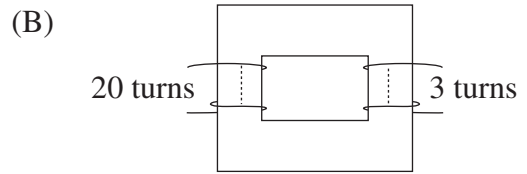
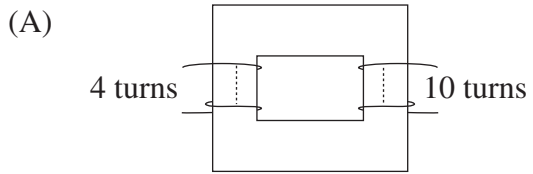


What is the current I ?

- (A) 1.5 A
(B) 1.7 A
(C) 3.0 A
(D) 6.0 A
- 9 Compared to a geostationary orbit, which row of the table correctly describes the relative properties of a low Earth orbit?

	<i>Orbital velocity</i>	<i>Orbital period</i>
(A)	Higher	Higher
(B)	Higher	Lower
(C)	Lower	Higher
(D)	Lower	Lower

- 10** Which of the following ideal transformers could be used to convert an input voltage of 20 volts AC to an output voltage of 2 volts AC?



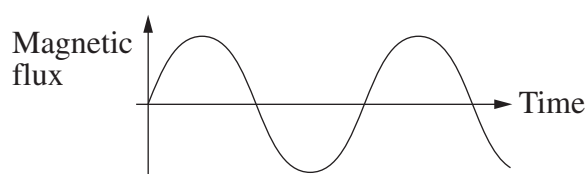
- 11** Which of the following is correct about the forces acting during a rocket launch?
- (A) Equal and opposite forces act on the rocket. This enables it to continue to accelerate even in the vacuum of space.
 - (B) The engines exert an upward thrust on the rocket. This thrust exceeds the downward force of the engines on the air.
 - (C) The rocket engines exert a downward force on the gases being expelled. These gases exert an upward force on the engines.
 - (D) The expelled gases exert a force against the launch pad. The launch pad then exerts an equal and opposite force on the rocket causing it to accelerate.

Use the data below to answer Questions 12 and 13.

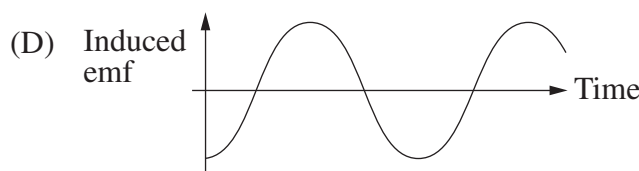
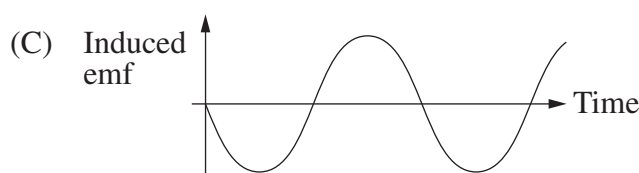
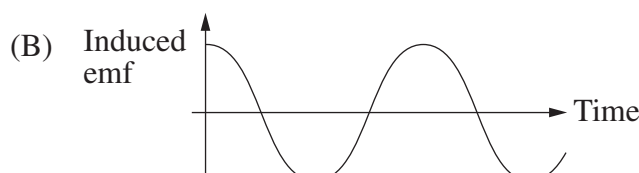
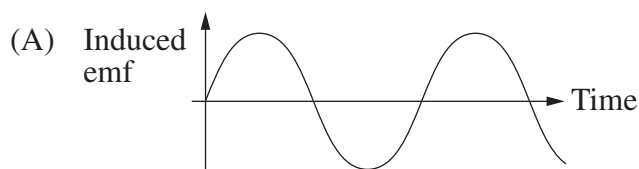
Orbital period of the Moon around Earth	$2.36 \times 10^6 \text{ s}$
Mean orbital radius of the Moon	$3.83 \times 10^8 \text{ m}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Mass of the Moon	$7.35 \times 10^{22} \text{ kg}$

- 12** What is the centripetal force experienced by the Moon due to Earth's influence?
- (A) $2.0 \times 10^{20} \text{ N}$
(B) $1.6 \times 10^{22} \text{ N}$
(C) $4.7 \times 10^{26} \text{ N}$
(D) $7.6 \times 10^{28} \text{ N}$
- 13** What is the orbital period of an Earth satellite having an orbital radius half that of the Moon?
- (A) $5.9 \times 10^5 \text{ s}$
(B) $8.3 \times 10^5 \text{ s}$
(C) $1.2 \times 10^6 \text{ s}$
(D) $7.5 \times 10^6 \text{ s}$

- 14 The graph shows variation in magnetic flux through a coil with time.



Which graph best represents the corresponding induced emf in the coil?



- 15 A magnet can be levitated above a superconductor, when the superconductor is below its critical temperature.

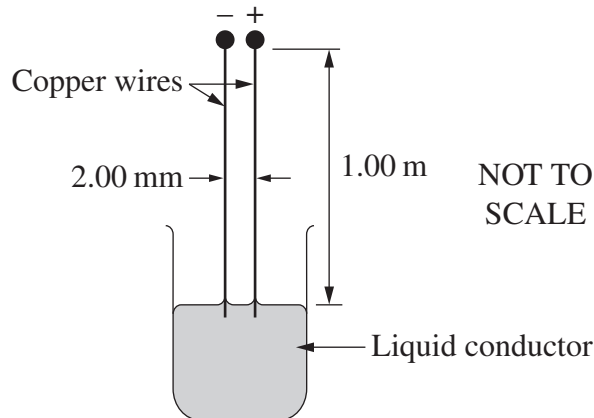
Which statement best describes how this occurs?

- (A) All of the magnet's field is absorbed by the superconductor.
- (B) The superconductor creates currents in the magnet, resulting in a repulsive force.
- (C) Cooper pairs are exchanged between the superconductor and the magnet through its field.
- (D) Some of the magnet's field is excluded by the superconductor, resulting in a repulsive force.

- 16 An ideal electric motor connected to a DC voltage source rotates at a constant rate of 200 revolutions per minute. There is no load on the motor.

Which of the following is a correct statement about the operation of the motor?

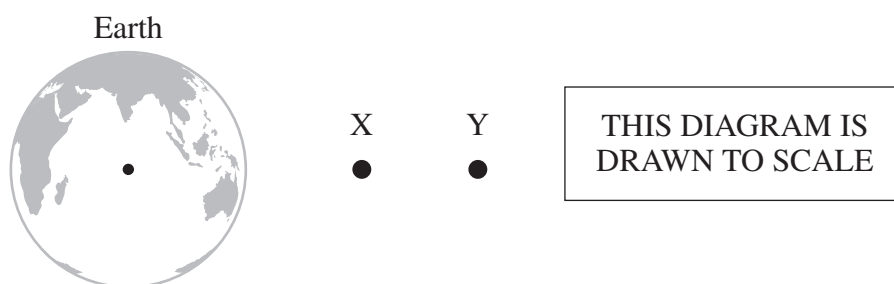
- (A) The applied voltage must exceed the back emf in order to keep the motor running.
 - (B) There is no back emf because it is only produced in AC motors due to the changing flux.
 - (C) The back emf is equal to the applied voltage because no work is being done by the motor.
 - (D) The back emf must exceed the applied voltage to prevent the motor's speed from increasing.
- 17 The following equipment is attached to a DC power supply.



What current must be flowing through the wires to result in a force of 2.50×10^{-3} N between them?

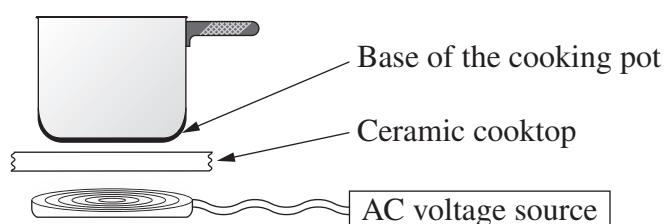
- (A) 0.224 A
- (B) 5.00 A
- (C) 12.5 A
- (D) 25.0 A

- 18 The gravitational force, due to Earth, on a mass positioned at X is F_x and on the same mass positioned at Y is F_y . The diagram is drawn to scale.



What is the value of $\frac{F_x}{F_y}$?

- (A) 1.5
 (B) 2.0
 (C) 2.25
 (D) 4.0
- 19 The diagram represents an induction cooking system.

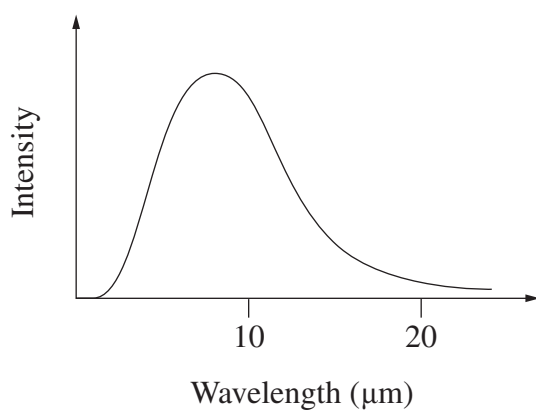


Which row in the following table shows the set of conditions that would result in the most rapid heating of the base of the cooking pot?

	<i>AC voltage frequency</i>	<i>Electrical resistance of pot base</i>
(A)	Low	Low
(B)	Low	High
(C)	High	Low
(D)	High	High

- 20 A company wishes to develop a semiconductor sensor to detect thermal radiation from people. The sensor will work on the same principle as a solar cell.

Typical Human Blackbody Radiation Curve



<i>Semiconductor</i>	<i>Band gap (eV)</i>
HgCdTe	0.03
InSb	0.17
Si	1.1
GaN	3.4

Using the information provided, which semiconductor would be most suitable for this purpose?

- (A) HgCdTe
- (B) InSb
- (C) Si
- (D) GaN

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Physics

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Centre Number

Section I (continued)

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Student Number

Part B – 55 marks

Attempt Questions 21–30

Allow about 1 hour and 40 minutes for this part

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Question 21 (6 marks)

- (a) Outline a first-hand investigation that could be performed to measure a value for acceleration due to gravity. 3

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- (b) How would you assess the accuracy of the result of the investigation? 1

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- (c) How would you increase the reliability of the data collected? 1

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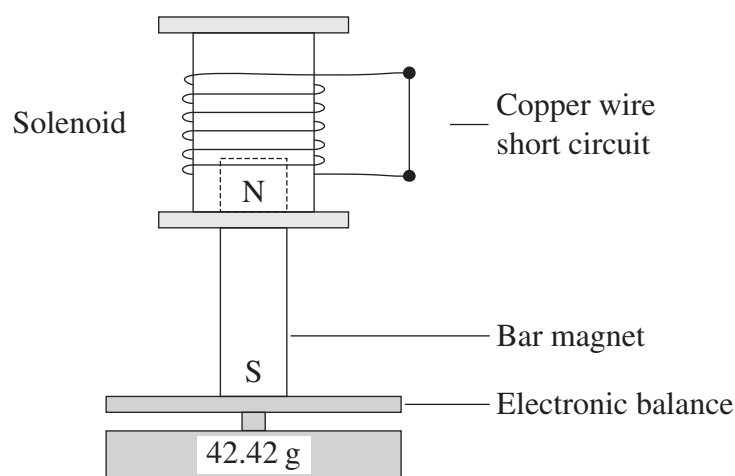
- (d) How would you assess the reliability of the data collected? 1

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Question 22 (6 marks)

A bar magnet is placed on a sensitive electronic balance as shown in the diagram. A hollow solenoid is held stationary, such that the magnet is partly within the solenoid.



The solenoid is then lifted straight up without touching the magnet. The reading on the balance is observed to change briefly.

- (a) Why does a current flow in the solenoid? 2

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- (b) Explain the reason for changes in the reading on the electronic balance as the solenoid is removed. 4

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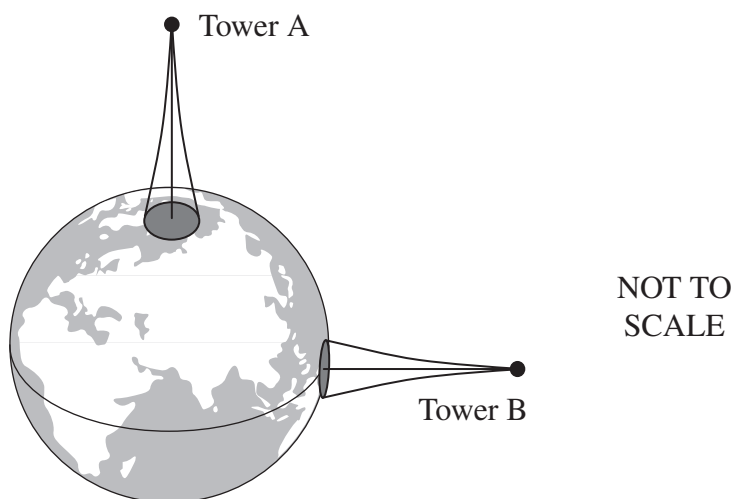
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Question 23 (4 marks)

Consider the following thought experiment.

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Two towers are built on Earth's surface. The height of each of the towers is equal to the altitude of a satellite in geostationary orbit about Earth. Tower A is built at Earth's North Pole and Tower B is built at the equator.



Identical masses are simultaneously released from rest from the top of each tower. Explain the motion of each of the masses after their release.

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Centre Number

Section I – Part B (continued)

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Student Number

Question 24 (5 marks)

- (a) Using a labelled diagram, outline the differences in the energy bands of conductors, semiconductors and insulators. **2**

- (b) Explain why a current is able to flow in a pure semiconductor when an electric field is applied across it. **3**

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Centre Number

Section I – Part B (continued)

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Student Number

Question 25 (7 marks)

- (a) Why have transistor circuits largely replaced thermionic devices? **2**

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- (b) Assess the impact of the use of transistors on society. **5**

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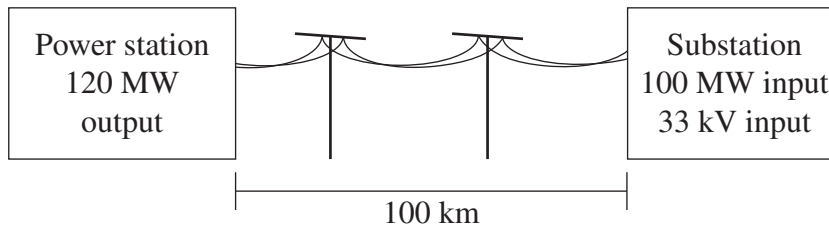
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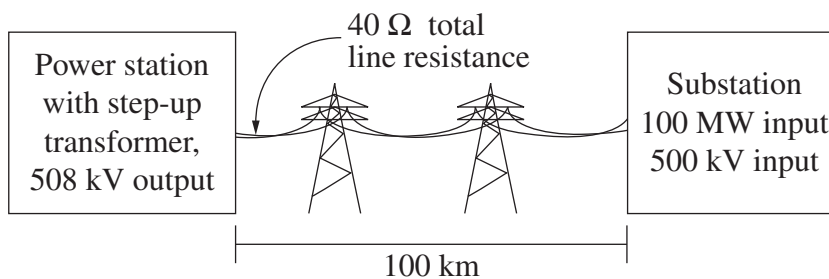
Question 26 (6 marks)

A transmission line is to be used to supply 100 MW of power from a power station to a substation 100 km away. Three possible designs are shown below.

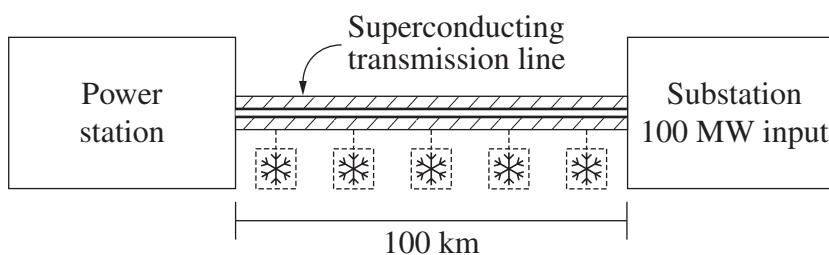
Design A – Low voltage



Design B – High voltage



Design C – Superconducting



 The cooling requirements for the superconducting transmission line consume 30 kW per km of line

Question 26 continues on page 21

Question 26 (continued)

Using physics principles and appropriate calculations, evaluate each design to determine the best way of transmitting power from the power station to the substation.

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End of Question 26

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Centre Number

Section I – Part B (continued)

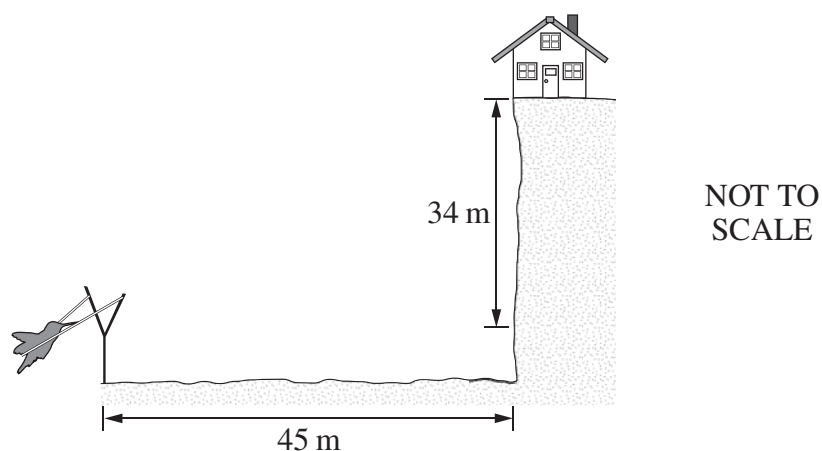
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Student Number

Question 27 (4 marks)

A toy bird is launched at 60° to the horizontal, from a point 45 m away from the base of a cliff.

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Calculate the magnitude of the required launch velocity such that the toy bird strikes the base of the wooden building at the top of the cliff, 34 m above the launch height.

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Centre Number

Section I – Part B (continued)

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Student Number

Question 28 (7 marks)

- (a) Outline ONE piece of evidence supporting Einstein's theory of relativity. **2**

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- (b) What criteria are used to test and validate a theory? **3**

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- (c) The distance between the cathode and screen in a cathode ray tube is 40 cm. **2**

If an electron travels through the tube at $3.0 \times 10^7 \text{ m s}^{-1}$, what is the apparent distance from the cathode to the screen in the electron's frame of reference?

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Centre Number

Section I – Part B (continued)

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Student Number

Question 29 (5 marks)

How is the motor effect used to produce different sounds in a loudspeaker? Include a labelled diagram.

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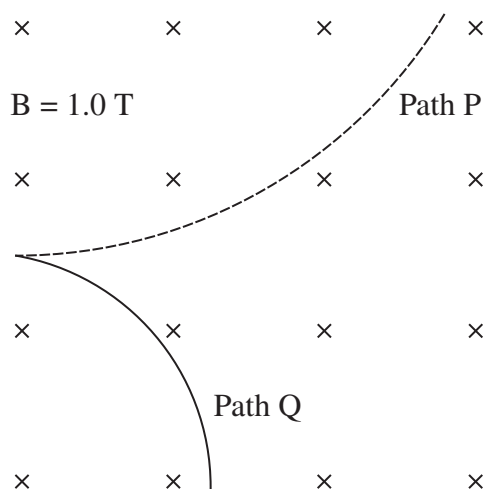
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Question 30 (5 marks)

The diagram shows the paths taken by two moving charged particles when they enter a region of uniform magnetic field.



- (a) Why do the paths curve in different directions? 1

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- (b) Why are the paths circular? 2

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- (c) How do the properties of a particle affect the radius of curvature of its path in a uniform magnetic field? 2

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Physics

Section II

25 marks

Attempt ONE question from Questions 31–35

Allow about 45 minutes for this section

For Questions 31–34

- answer parts (a)–(c) of the question in Section II Answer Booklet 1.
- answer parts (d)–(e) of the question in Section II Answer Booklet 2.

Extra writing booklets are available.

For Question 35

- answer parts (a)–(b) of the question in Section II Answer Booklet 1.
- answer parts (c)–(d) of the question in Section II Answer Booklet 2.

Extra writing booklets are available.

Show all relevant working in questions involving calculations.

	Pages
Question 31 Geophysics	31–33
Question 32 Medical Physics	34
Question 33 Astrophysics	35–36
Question 34 From Quanta to Quarks	37
Question 35 The Age of Silicon	38–39

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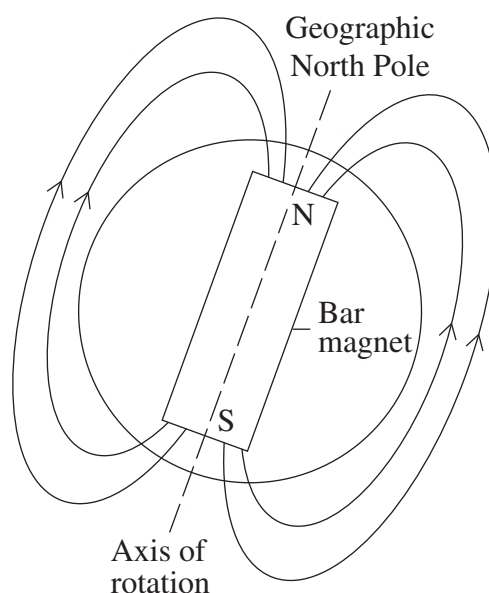
Question 31 — Geophysics (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) Jean Richer observed that a pendulum clock, which was accurately calibrated in Paris, lost about 150 seconds per day when it was used on the equatorial island of Cayenne. **3**

Outline the effects of TWO variables that changed the behaviour of the clock.

- (b) (i) Why does Earth's magnetic field change over time? **2**
- (ii) A website used the following model to explain Earth's magnetic field. **3**

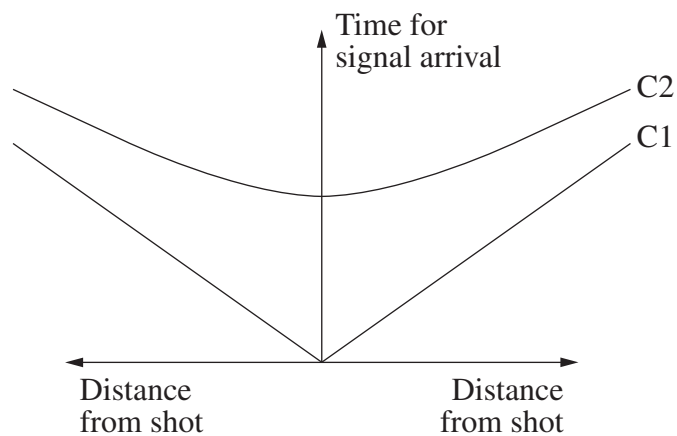


Analyse the validity of this model.

Question 31 continues on page 32

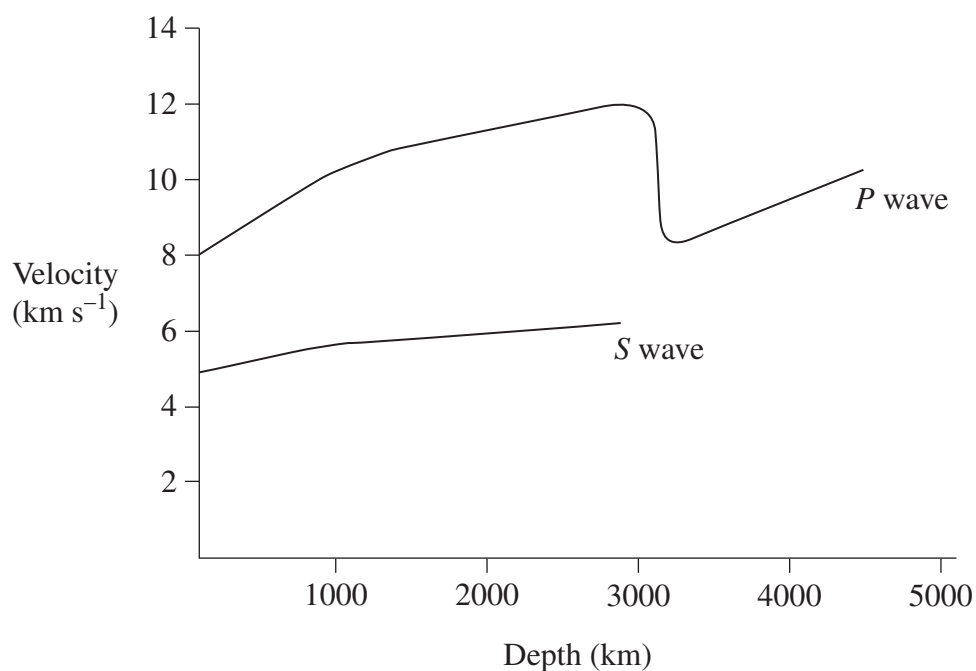
Question 31 (continued)

- (c) (i) Data obtained from a shot-geophone analysis is shown on the following graph. 2



Outline what the lines C1 and C2 on this graph represent.

- (ii) Account for the behaviour of *P* waves and *S* waves represented by the following graph. 4



Question 31 continues on page 33

Question 31 (continued)

Answer parts (d)–(e) in Section II Answer Booklet 2.

- | | | | |
|-----|------|--|----------|
| (d) | (i) | What is the significance of the magnetic anomalies associated with rocks adjacent to mid-ocean ridges? | 3 |
| | (ii) | Other than the geometric fit of continents or magnetic evidence, what evidence supports the theory of plate tectonics? | 2 |
| (e) | | How is remote sensing used to obtain information about Earth? | 6 |

End of Question 31

Question 32 — Medical Physics (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

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|-----|-------|---|----------|
| (a) | (i) | The first X-ray image of a human was made in 1895 but it was not until 1972 that CAT scan imaging was developed. | 2 |
| | | Account for the time interval between the development of these two related technologies. | |
| | (ii) | Contrast the information provided by PET scans and CAT scans. | 2 |
| | (iii) | Contrast the production of radiation used for PET scans and CAT scans. | 3 |
| (b) | (i) | What effects do the pulses of radio waves have on hydrogen nuclei in the body of a person having a magnetic resonance image scan? | 3 |
| | (ii) | Outline how advances in physics in the 20th century contributed to the development of magnetic resonance imaging. | 3 |
| (c) | | Describe the properties that make a radioactive isotope useful for medical imaging. Include a specific example of a radioactive isotope in your answer. | 3 |

Answer parts (d)–(e) in Section II Answer Booklet 2.

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|-----|--|----------|
| (d) | Explain how endoscopes obtain images of internal organs. | 3 |
| (e) | Discuss the range of information obtained from, and the limitations of, ultrasound medical imaging technology. | 6 |

Question 33 — Astrophysics (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

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|-----|------|--|----------|
| (a) | (i) | Why do astronomers use both apparent and absolute measurements in photometry? | 1 |
| | (ii) | Compare how astronomers obtain photometric and spectroscopic data. | 4 |
| (b) | (i) | Describe the problems associated with ground-based optical astronomy in terms of resolution. | 3 |
| | (ii) | How can the resolution of ground-based optical telescopes be improved? | 2 |
| (c) | (i) | Draw a flow diagram for stellar evolution using the following celestial objects: <ul style="list-style-type: none">• Black hole• Main sequence star• Neutron star/pulsar• Planetary nebula• Red giant• Supernova• White dwarf. | 2 |
| | (ii) | All naturally occurring elements have been synthesised within stars. Explain how this occurs. | 3 |

Answer parts (d)–(e) in Section II Answer Booklet 2.

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|-----|------|---|----------|
| (d) | (i) | Explain how trigonometric parallax is used to determine the distances to stars. | 3 |
| | (ii) | Outline ONE limitation of using trigonometric parallax to determine distances to stars. | 1 |

Question 33 continues on page 36

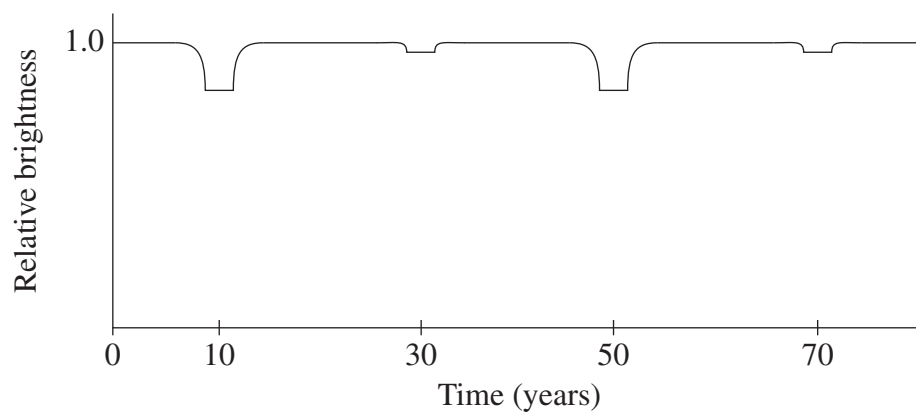
Question 33 (continued)

- (e) Astronomers have obtained the following data about stars in a binary system.

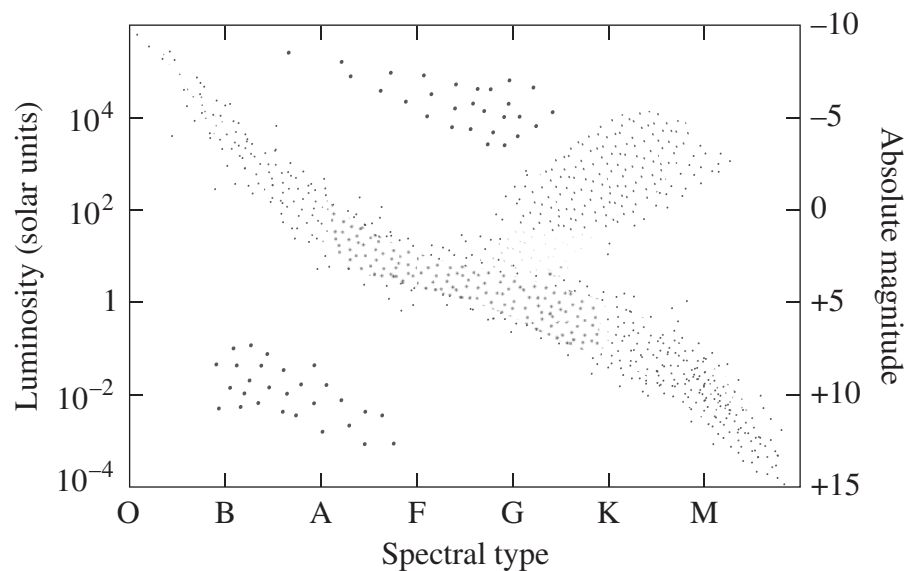
6

	<i>Star A</i>	<i>Star B</i>
Apparent magnitude	0.34	13.16
Spectral class	F5	A4
Luminosity (relative to Sun)	7.3	0.0063
Average separation	2.2×10^9 km	

Light Curve for the Binary System



Hertzsprung-Russell Diagram



Analyse the above data to write a detailed description of this binary system.

End of Question 33

Question 34 — From Quanta to Quarks (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- | | | | |
|-----|------|---|----------|
| (a) | (i) | Using diagrams and text, outline Bohr’s explanation of the Balmer series. | 3 |
| | (ii) | Outline TWO observations from atomic emission spectra that could not be fully explained by the Rutherford-Bohr model. | 2 |
| (b) | (i) | What are the requirements for an uncontrolled nuclear chain reaction? | 3 |
| | (ii) | How does the equivalence between mass and energy relate to what occurs during the natural radioactivity process? | 2 |
| (c) | | How can neutrons be used to probe the positions of nuclei within the structure of a metal crystal? Make reference to the work of Louis de Broglie in your answer. | 4 |

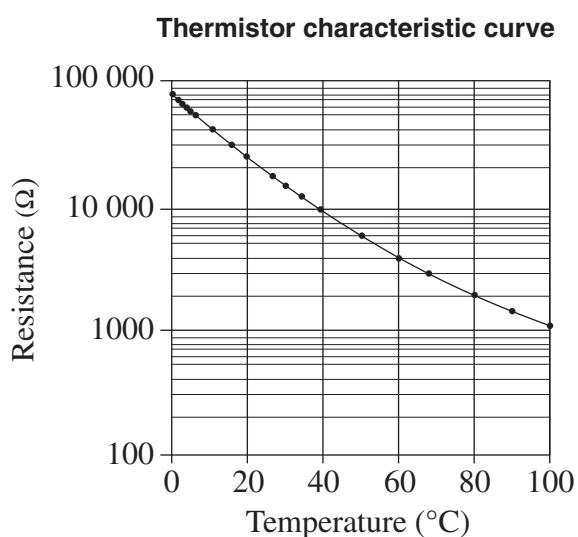
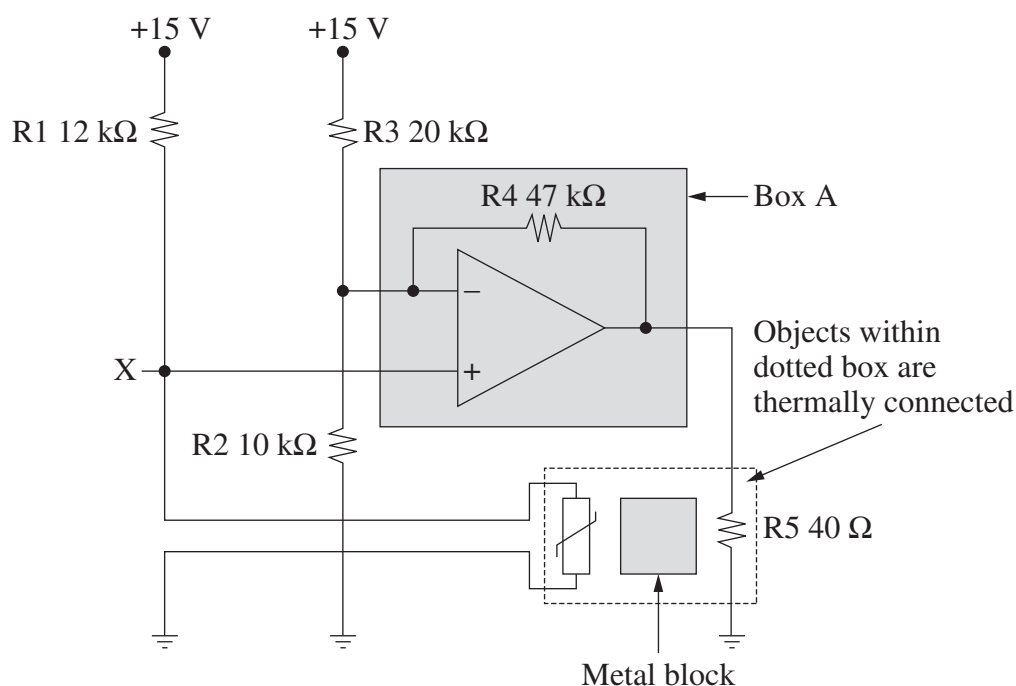
Answer parts (d)–(e) in Section II Answer Booklet 2.

- | | | | |
|-----|------|---|----------|
| (d) | (i) | Explain how the internal structures of the proton and neutron determine their properties. | 3 |
| | (ii) | How are accelerators used to investigate the structure of matter? | 2 |
| (e) | | Describe the role of conservation laws in the development of atomic physics. | 6 |

Question 35 — The Age of Silicon (25 marks)

Answer parts (a)–(b) in Section II Answer Booklet 1.

- (a) A circuit and a graph of the thermistor characteristic curve are shown. In the circuit the thermistor and R5 are firmly attached to a metal block, and all three are at the same temperature.



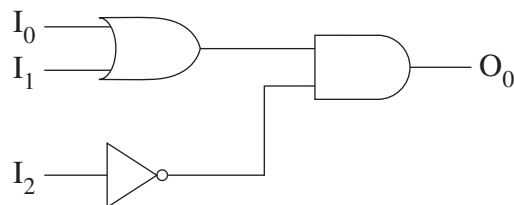
Question 35 continues on page 39

Question 35 (continued)

- (i) Classify the circuit on the opposite page as an electric circuit or an electronic circuit and justify your choice. **2**
 - (ii) Identify the functions of the TWO transducers in this circuit. **2**
 - (iii) Draw a set of axes in your booklet and sketch how the voltage at point X varies with temperature. **3**
 - (iv) Describe the function of the circuit in Box A. **4**
 - (v) Calculate the equilibrium temperature achieved by the metal block in this circuit, assuming that the block is thermally connected to the thermistor and R5. **3**
- (b) Explain the use of relays, rather than transistors, to switch large currents in circuits. **2**

Answer parts (c)–(d) in Section II Answer Booklet 2.

- (c) For the logic circuit shown, construct a truth table. **3**



- (d) Analyse how the development of modern electronics has been influenced by our increased understanding of the properties of materials and the development of complex manufacturing techniques. **6**

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DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi} \right)$	$2.0 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{\text{av}} = \frac{\Delta r}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} \text{ therefore } a_{\text{av}} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2}a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = - \frac{R_{\text{f}}}{R_{\text{i}}}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

[illegible]

Lanthanoids

57	La	138.9
58	Ce	140.1
59	Pr	140.9
60	Nd	144.2
61	Pm	
62	Sm	150.4
63	Eu	152.0
64	Gd	157.3
65	Tb	158.9
66	Dy	162.5
67	Ho	164.9
68	Er	167.3
69	Tm	168.9
70	Yb	173.1
71	Lu	175.0
	Lutetium	

Actinoids

89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr
Actinium	Thorium	Protactinium	Uranium	Neptunium	Plutonium	Americium	Curium	Berkelium	Californium	Einsteinium	Fermium	Mendelevium	Nobelium	Lawrencium

Elements with atomic numbers 113 and above have been reported but not fully authenticated.

Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of data. Some data may have been modified.



2012 HSC Physics Marking Guidelines

Section I, Part A

Multiple-choice Answer Key

Question	Answer
1	D
2	A
3	C
4	B
5	C
6	C
7	B
8	A
9	B
10	D
11	C
12	A
13	B
14	D
15	D
16	C
17	B
18	C
19	C
20	B

Section I, Part B

Question 21 (a)

Criteria	Marks
- Provides details of the method used, including what data would be collected and repetition for reliability - Outlines how data would be used to determine the acceleration due to gravity	3
Provides details of the method used, including what data would be collected.	2
Identifies a practical method that could lead to a determination of 'g'	1

Question 21 (b)

Criteria	Marks
Indicates that the value obtained experimentally should be compared with an accepted standard value obtained from a reliable source.	1

Question 21 (c)

Criteria	Marks
States that repetition of the experiment is required to establish reliability	1

Question 21 (d)

Criteria	Marks
States that the less the measurements vary from the mean, the more reliable is the experimental data.	1

Question 22 (a)

Criteria	Marks
- Relates a change in flux through the coil when it is moved, to the production of an emf across the coil - Indicates that a current flows because the short circuit completes the current pathway through the coil	2
States that there is a change in flux through the coil when it is moved	1

Question 22 (b)

Criteria	Marks
- Relates the correct polarity of the coil's field to the change in flux through the coil - Relates the coil's magnetic field to the production of a force on the magnet that causes a reduction in the reading on the balance - Relates the increasing distance between the magnet and the coil to the return of the reading on the balance to its original value before the coil was moved	4
- Relates the change in flux through the coil to the correct polarity of the coil's field - Relates the coil's magnetic field to the production of a force on the magnet that causes a reduction in the reading on the balance	3
Relates that the change in the reading on the balance is due to a force caused by the current in the coil.	2
Any relevant statement	1

Question 23

Criteria	Marks
- Describes the behaviour of the mass dropped from tower B and relates this to the motion of tower B - Describes the path and change in acceleration of the mass dropped from tower A and relates this to the location of tower A and the change in acceleration with decreasing distance from Earth's centre.	4
- Describes the behaviour of the mass dropped from tower B and relates this to the motion of tower B - Describes the path of the mass dropped from tower A and relates this to the location of tower A.	3
- Relates the motion of either mass to the location of the tower OR - Describes the path of both masses	2
Provides one relevant piece of information	1

Question 24 (a)

Criteria	Marks
- Correctly labelled diagram - Shows relative band gaps for all three materials.	2
Evidence of an understanding of the existence of a band gap in materials	1

Question 24 (b)

Criteria	Marks
Relates the formation of electron-hole pairs and the movement of carriers in the valence and conduction bands to the applied electric field.	3
- Relates the movement of electrons or holes to the applied electric field OR - Describes the formation of electron-hole pairs in the conduction-valence bands	2
Any correct statement about the current in a semiconductor	1

Question 25 (a)

Criteria	Marks
Identifies two differences between transistors and thermionic devices	2
Identifies one difference between transistor circuits and thermionic devices	1

Question 25 (b)

Criteria	Marks
Makes a judgment, supported by detailed evidence, of the effects of transistors. Effects described must relate explicitly to society. The answer reflects depth of understanding, evidenced by the use of scientific language and thorough understanding of the connection between transistors and changes in society	5
Makes a judgment, supported by evidence, of the effects of transistors on society. The answer uses some scientific language and demonstrates an understanding of the connection between transistors and changes in society	4
Makes a judgment, identifies an application of transistors which affect either individual people, or society	3
Identifies an application of transistors, and an associated effect of transistors, either on individual people, or on society	2
Identifies an application of transistors or an effect of transistors on people	1

Question 26

Criteria	Marks
- Calculates the power loss resulting from all three designs and concludes, based on the least power lost/consumed that design B is the best of the three systems. Answer shows all calculations with correct and comparable units. - Shows a clear understanding of the physics principles involved	6
- Calculates the power usage/loss in two systems, including correct scientific units AND - Applies a correct method for the third (but does not arrive at the correct answer with units) OR - Makes a correct judgment of the best design	5
- Calculates the power usage/loss in design B and one of A or C OR - Relates the principles of power loss to the three designs	4
- Calculates the power usage for two designs OR - Relates the principles of power loss to two designs	3
- Calculates the power usage of one design OR - Relates the principles of power loss to the design	2
A relevant statement that demonstrates understanding of the problem	1

Question 27

Criteria	Marks
• Correctly substitutes into the derived equation	4
• Equates derived expressions	3
• Derives correct expressions for time of flight	2
• Derives correct expressions for horizontal and vertical components of launch velocity	1

Question 28 (a)

Criteria	Marks
• Briefly describes an experimental result or technological application, which is consistent with one prediction of the theory of relativity.	2
• Identifies relevant evidence	1

Question 28 (b)

Criteria	Marks
• Shows a deep understanding of the process of validating a theory based on agreement with observations and experiments and the capacity to make correct predictions	3
• Shows a limited understanding of the process of validating a theory	2
• Demonstrates an elementary understanding of how a theory is validated eg by stating one way that a theory can be validated	1

Question 28 (c)

Criteria	Marks
• Correct formula with correct substitution	2
• Substitutes one correct value into the length contraction formula	1

Question 29

Criteria	Marks
• Draws a labelled diagram showing the coil, magnet and speaker cone with the wire in the coil cutting across the magnetic field • Describes the oscillation of the coil/cone produced by AC in the coil in a magnetic field (motor effect) and states that the vibration of the cone produces the sound • Accounts for the production of louder sounds through the use of larger currents in the coil • Relates the pitch (frequency) of sound to the changes in the current through the speaker coil	5
• Draws a labelled diagram showing the coil and the magnet with the wire in the coil within the magnetic field • Describes the oscillation of the coil/cone produced by AC/varying current in the coil in a magnetic field (motor effect) • Relates changes in both loudness OR frequency of the sound to changes in the current through the coil	4
• Draws a labelled diagram showing a wire near a magnet • Describes the force on the wire produced by the current in a magnetic field (motor effect) • Identifies that the vibration of the coil/cone produces a sound	3
• Draws a labelled diagram AND • Describes the force on the wire produced by the current in a magnetic field (motor effect) OR • Draws a labelled diagram showing a wire and a magnet OR • States that an alternating current/ varying current passes through the wire OR • Identifies that the vibration of the coil/cone produces sound	2
• Any relevant statement OR • Diagram showing a coil and a magnet	1

Question 30 (a)

Criteria	Marks
• States that the two particles have opposite charges	1

Question 30 (b)

Criteria	Marks
• Provides reasons	2
• Provides a reason	1

Question 30 (c)

Criteria	Marks
• Qualitatively relates the radius of curvature of the path to two properties of the particle	2
• Relates the radius of curvature to one property of the particle	1

Section II

Question 31 – Geophysics

Question 31 (a)

Criteria	Marks
Relates two variables to the effect that they produced on the pendulum's period	3
- Identifies two variables that changed the period of the pendulum OR - Relates a variable to the effect that it produced on the pendulum's period	2
Identifies one variable that changed the period of the pendulum	1

Question 31 (b) (i)

Criteria	Marks
Describes one variable that changes over time and how it affects the field	2
Identifies one changing variable that affects the Earth's magnetic field	1

Question 31 (b) (ii)

Criteria	Marks
Shows how parts of the model fail to relate to observations of the Earth's magnetic properties or the scientific representation of magnetic fields	3
- Identifies two deficiencies in the model OR - Explains one deficiency	2
Identifies a deficiency in the model	1

Question 31 (c) (i)

Criteria	Marks
• Outlines what C1 and C2 represent	2
• Makes a correct statement about C1 or C2	1

Question 31 (c) (ii)

Criteria	Marks
• Features of both graphs are related to Earth's internal structure	4
• Accounts for features of one graph and a feature of the other in terms of the Earth's structure	3
• Accounts for features of one graph OR a feature of the other in terms of the Earth's structure	2
• Relates the Earth's structure to a feature of either graph	1

Question 31 (d) (i)

Criteria	Marks
• Relates the magnetic anomaly pattern to processes taking place at the plate boundary and its significance to the theory of plate tectonics	3
• Describes seafloor spreading that is taking place at the mid-ocean ridge and relates this to the magnetic anomalies	2
• Describes a feature of the magnetic anomaly pattern OR • Makes a correct statement about divergent plate boundaries	1

Question 31 (d) (ii)

Criteria	Marks
• Relates plate tectonic theory to evidence supporting it	2
• Identifies evidence consistent with plate tectonic theory	1

Question 31 (e)

Criteria	Marks
Explains a range of remote sensing processes and the physical principles upon which they depend. Uses scientific language to clarify the response	6
Describes a range of remote sensing processes and at least one physical principle upon which one of these depends	4–5
Identifies examples of remote sensing and mentions a physics principle	2–3
Identifies an example of remote sensing	1

Question 32 – Medical Physics

Question 32 (a) (i)

Criteria	Marks
- Displays a clear understanding of the fact that CAT scans produce more data than X-rays - Outlines that a named technology for CAT was not readily available until 1972	2
- Shows some understanding of the fact that CAT produces more data than X-rays OR - Shows understanding that a named technology required for CAT scans was not available before 1972	1

Question 32 (a) (ii)

Criteria	Marks
Displays a clear understanding of the differences in the information obtained using each technique	2
Shows understanding of the information obtained using one technique	1

Question 32 (a) (iii)

Criteria	Marks
Shows clear understanding of the differences in the production mechanism of radiation used in both technologies	3
- Shows some understanding of the production of X-ray radiation for CAT AND - Shows clear understanding of the production of gamma rays for PET OR - Shows some understanding of the production of gamma rays for PET AND - Shows clear understanding of the production of X-rays for CAT	2
- Shows some understanding of the production of X-ray radiation for CAT OR - Shows clear understanding of the production of gamma rays for PET	1

Question 32 (b) (i)

Criteria	Marks
Shows a clear link of the effects of radio waves on the hydrogen nuclei's spin alignment	3
Shows some understanding of the effect that radio waves have on the alignment and/or precession of proton nuclei	1–2

Question 32 (b) (ii)

Criteria	Marks
Shows a clear understanding of the technologies involved with MRI scans and provides clear outlines of the advances in physics understanding required to develop these	3
Shows some understanding of the technologies involved with MRI scans and provides a basic connection to the advances in physics understanding associated with these	2
- Shows some understanding of the technologies involved with MRI scans OR - Shows some understanding of the physics development that was required to develop an MRI device	1

Question 32 (c)

Criteria	Marks
- Clearly describes at least two desirable properties of radioactive isotopes for medical imaging AND - Provides a named example of a radioactive isotope with medical application	3
- States desirable properties of a radioactive isotope for medical applications OR - Names example of a radioactive isotope with medical applications and states one property of the isotope	2
- Shows some understanding of the desirable properties of a radioactive isotope for medical applications OR - Provides a named example of a radioactive isotope with medical applications	1

Question 32 (d)

Criteria	Marks
Shows a clear understanding of how endoscopes work, including the function and design of the optical fibres	3
Shows some understanding of the types of fibre bundles used in endoscopes but is unclear about their specific function or design	2
- Shows some understanding of the fibre bundles used in endoscopes OR - Shows some understanding of how optical fibres work	1

Question 32 (e)

Criteria	Marks
Uses scientific language	6
Discusses multiple uses and limitations of medical ultrasound imaging	4–5
Describes some uses and limitations of ultrasound imaging	2–3
Outlines uses and/or limitations of ultrasound imaging	1
Identifies a use or a limitation of ultrasound imaging	

Question 33 – Astrophysics

Question 33 (a) (i)

Criteria	Marks
Identifies a reason	1

Question 33 (a) (ii)

Criteria	Marks
Compares methods used to obtain both photometric and spectroscopic data	4
Describes methods used to obtain both photometric and spectroscopic data	3
Identifies a method by which each of photometric and spectroscopic data are obtained by astronomers	2
Identifies one method by which photometric or spectroscopic data about stars is obtained	1

Question 33 (b) (i)

Criteria	Marks
Relates limitations in resolution of ground-based optical telescopes to a cause of their limitations	3
- Identifies a factor which limits resolution of optical telescopes - Relates the limit to its cause	2
Identifies a factor that limits the resolution of a telescope	1

Question 33 (b) (ii)

Criteria	Marks
Describes methods of improving the resolution of optical telescopes	2
Identifies one method of improving resolution of an optical telescope	1

Question 33 (c) (i)

Criteria	Marks
Flow diagram correctly relates all of the identified objects	2
Flow diagram correctly relates three of the identified objects	1

Question 33 (c) (ii)

Criteria	Marks
• Describes conditions and processes of hydrogen to helium conversion, helium burning and heavy element nucleosynthesis in supernovae	3
• Outlines two processes by which heavy elements are produced	2
• Identifies a process by which lighter nuclei form heavier nuclei	1

Question 33 (d) (i)

Criteria	Marks
• Relates the calculated distance to the motion of the Earth around the Sun and the relatively fixed, distant stars	3
• Shows a basic understanding of how parallax shift of a star can, due to Earth's orbital motion, be used to calculate the star's distance	2
• Identifies a variable that is significant in the process of determining the distance to a star using parallax	1

Question 33 (d) (ii)

Criteria	Marks
• Outlines a limitation to the use of the parallax method of determining stellar distance	1

Question 33 (e)

Criteria	Marks
• Gives a detailed description of the system including quantitative analysis of at least TWO characteristics • Gives detailed description of each star comparing characteristics (from Hertzsprung-Russell diagram) • Uses scientific language	6
• Gives a description of the system including a quantitative analysis • Gives a description of a star's characteristics • Uses scientific language	4–5
• Gives a description of the system based on analysis OR • Identifies the stars in the system based on analysis	2–3
• Any correct statement drawn from the data	1

Question 34 – From Quanta to Quarks

Question 34 (a) (i)

Criteria	Marks
- Shows a clear understanding of Bohr's model - Shows a clear understanding of how the Balmer series is produced - Uses a clear labelled diagram to aid in explaining either Bohr's model OR the Balmer series	3
- Shows some understanding of the Bohr model - Shows some understanding of the Balmer series - Uses a diagram to aid their explanation	2
- Shows some understanding of the Bohr model OR - Shows some understanding of the Balmer series	1

Question 34 (a) (ii)

Criteria	Marks
Clearly outlines TWO observations	2
States at least one observation	1

Question 34 (b) (i)

Criteria	Marks
Shows a clear understanding of the key requirements	3
Shows a clear understanding of the majority of the key requirements	2
Shows a clear understanding of at least one of the key requirements	1

Question 34 (b) (ii)

Criteria	Marks
Shows a good understanding of the relevance of Einstein's $E=mc^2$ to the radioactive decay process	2
- Shows some understanding of the connection between Einstein's $E=mc^2$ and radioactive decay OR - Shows some understanding of the mass changes during radioactive decay	1

Question 34 (c)

Criteria	Marks
- Shows a clear understanding of the wave nature of moving neutrons - Shows a clear understanding that the periodic arrangement of metal nuclei will diffract the neutrons in turn revealing their arrangement - Shows a clear understanding that the neutral neutrons will not scatter from the electrons - Connects the above points into a coherent description	4
Shows some understanding of the above points and some understanding of how they are related	3
Shows some understanding of the above points	1–2

Question 34 (d) (i)

Criteria	Marks
- Outlines the number and types of quarks in the proton and the neutron - Identifies two composite properties and outlines how these are determined by the quarks present in each case	3
- Shows some understanding of the internal structure of protons and neutrons - Identifies a composite property and outlines how this is determined by the internal structure	2
Shows some understanding of the internal structure of the proton or neutron	1

Question 34 (d) (ii)

Criteria	Marks
- Identifies charged particles are accelerated - Identifies that collision products are analysed	2
- Identifies charged particles are accelerated OR - Identifies that collision products are analysed	1

Question 34 (e)

Criteria	Marks
- Displays a good understanding of the role that conservation laws have played in the development of atomic physics - Outlines more than one relevant example of a conservation law and relates them to developments in atomic physics - Provides a coherent description and uses scientifically correct terminology throughout	5–6
- Displays an understanding of the role the conservation laws have played in the development of atomic physics - Outlines one relevant example of a conservation law and relates it to developments in atomic physics	3–4
- Displays some understanding of the role that conservation laws have played in the development of atomic physics OR - Outlines ONE example of a conservation law relevant to atomic physics	1–2

Question 35 – The Age of Silicon

Question 35 (a) (i)

Criteria	Marks
- Correctly classifies the circuit - Provides valid reasoning	2
Correct statement related to the classification of the circuit	1

Question 35 (a) (ii)

Criteria	Marks
- Identifies the two transducers - Identifies the function of the two transducers	2
- Identifies a transducer - Identifies its function	1

Question 35 (a) (iii)

Criteria	Marks
Draws a correctly labelled graph with the correct line plotted using the potential divider formula	3
- Draws a correctly labelled graph with clearly incorrect values plotted but with the correct general trend OR - Plots the correct values but with the graph incorrectly formatted / labelled	2
Displays understanding of the need to read values from the provided plot and use the potential divider formula to find the potentials	1

Question 35 (a) (iv)

Criteria	Marks
- Outlines the configuration of the circuit in Section A - Relates this configuration to the operation of the circuit	3–4
Displays some understanding of the configuration or operation of the circuit	1–2

Question 35 (a) (v)

Criteria	Marks
- Calculates an equilibrium temperature - Demonstrates an understanding of the effect of R4 on the operation of the circuit	3
Calculates an equilibrium temperature but ignores R4	2
- Displays a clear understanding of how to perform the calculation but does not arrive at the correct result OR - Displays a clear understanding of the operation of the circuit but does not arrive at a correct result	1

Question 35 (b)

Criteria	Marks
Displays a clear understanding of when to use each and why	2
Displays some understanding of the relevance of transistors and relays to high current switching	1

Question 35 (c)

Criteria	Marks
Correctly calculates the truth table	3
Correctly constructs a truth table with the majority of entries correct	1 – 2

Question 35 (d)

Criteria	Marks
• Demonstrates a good overall understanding of the relationship between the understanding of the properties of materials and the development of modern electronics • Demonstrates a good overall understanding of the relationship between the development of complex manufacturing techniques and the development of modern electronics • Shows an understanding of the relationship between the above two points • Uses clear, concise and scientifically correct language	5–6
• Demonstrates some understanding of the relationship between the understanding of the properties of materials and the development of modern electronics • Demonstrates some understanding of the relationship between the development of complex manufacturing techniques and the development of modern electronics • Shows some understanding of the relationship between the above two points	3–4
• Demonstrates limited understanding of the relationship between the understanding of the properties of materials and the development of modern electronics AND/OR • Demonstrates limited understanding of the relationship between the development of complex manufacturing techniques and the development of modern electronics	1–2

Physics

2012 HSC Examination Mapping Grid

Section I Part A

Question	Marks	Content	Syllabus outcomes
1	1	9.3.3.2.3, 9.3.1.2.6, 9.3.5.2.1	H9
2	1	9.2.4.2.8	H3
3	1	9.4.1.2.9	H9, H10
4	1	9.2.1.2.3	H12.3c, H9
5	1	9.4.3.2.6, 7	H3
6	1	9.4.1.2.6	H9, H13.1e
7	1	9.3.1.3.5	H9
8	1	9.3.1.3.3	H9
9	1	9.2.2.2.9	H9
10	1	9.3.4.3.2	H9, H7, H12.3c
11	1	9.2.2.2.7	H9
12	1	9.2.2.3.4	H9
13	1	9.2.2.3.5	H9
14	1	9.3.2.2.4, 9.3.2.2.5	H9
15	1	9.4.4.3.3	H9
16	1	9.3.2.2.6	H9
17	1	9.3.1.3.1, 9.3.1.2.2	H9
18	1	9.2.3.3.2	H9
19	1	9.3.2.3.3	H9
20	1	9.4.3.3.4, 9.4.2.3.4, 9.4.3.2.2, 9.4.2.2.6	H3

Section I Part B

Question	Marks	Content	Syllabus outcomes
21 (a)	3	9.2.1.3.1	H11
21 (b)	1	9.1	H11.2c, H12, H12.4c
21 (c)	1	9.1	H11.2c, H12, H12.2b
21 (d)	1	9.1	H11.2c, H12, H12.4e
22 (a)	2	9.3.2.3.1, 9.3.2.3.2, 9.3.2.2.4	H9
22 (b)	4	9.3.2.3.1, 9.3.2.3.2, 9.3.2.2.4, 9.2.1.3.3	H9
23	4	9.2.2.2.9, 9.2.3.3.1, 9.2.2.2.8	H9
24 (a)	2	9.4.3.2.2	H13.1e
24 (b)	3	9.4.3.3.1, 9.4.3.2.4, 9.4.3.2.3	H9
25 (a)	2	9.4.3.2.8, 9.4.3.3.3	H3
25 (b)	5	9.4.1, 9.4.3.3.2	H4

Question	Marks	Content	Syllabus outcomes
26	6	9.3.3.2.4, 9.3.4.3.4, 9.1, 9.4.4.3.5	H3, H7, H13.1e, H14.1c, H14.1f, H14.2b, H14.3c, H14.2d
27	4	9.2.2.3.1	H9
28 (a)	2	9.2.4.3.4	H6
28 (b)	3	9.2.4.3.4	H2
28 (c)	2	9.2.4.3.5	H6
29	5	9.3.1.3.5	H9
30 (a)	1	9.4.1.2.5, 9.4.1.2.3	H9
30 (b)	2	9.4.1.3.3, 9.2.2.3.4	H9
30 (c)	2	9.4.1.3.3, 9.2.2.3.4, 9.2.2.2.7	H9

Section II

Question	Marks	Content	Syllabus outcomes
Question 31		Geophysics	
31 (a)	3	9.5.1.3.1, 9.5.2.2.4, 9.5.2.3.2	H9
31 (b) (i)	2	9.5.4.2.2	H9
31 (b) (ii)	3	9.5.4.2.1, 9.5.4.3.1	H9
31 (c) (i)	2	9.5.3.3.2	H8
31 (ii)	4	9.5.3.2.1, 9.5.3.2.2, 9.5.3.2.3, 9.5.3.3.1	H8
31 (d) (i)	3	9.5.4.2.2, 9.5.4.2.3, 9.5.4.3.2	H9
31 (ii)	2	9.5.4.2.3, 9.5.4.2.4	H9
31 (e)	6	9.5.1.1, 9.5.2.2.2, 9.5.2.2.3	H8, H10
Question 32		Medical Physics	
32 (a) (i)	2	9.6.2.2.3, 9.6.2.2.1	H10
32 (a) (ii)	2	9.6.2.2.4, 9.6.2.3.2, 9.6.3.2.5, 9.6.3.3.1	H10
32 (a) (iii)	3	9.2.2.1, 9.6.3.2.1, 9.6.3.2.4	H10
32 (b) (i)	3	9.6.4.2.6	H9, H10
32 (b) (ii)	3	9.6.4.3.3, 9.6.4.3.5	H3
32 (c)	3	9.6.3.2.1, 9.6.3.2.2	H3
32 (d)	3	9.6.2.2.5-7, 9.6.2.3.3	H10
32 (e)	6	9.6.1.1	H3, H8

Question 33		Astrophysics	
33 (a) (i)	1	9.7.4.2.1	H10
33 (a) (ii)	4	9.7.4.3.2, 9.7.3.2.2	H10
33 (b) (i)	3	9.7.1.2.4	H10
33 (b) (ii)	2	9.7.1.2.5	H3
33 (c) (i)	2	9.7.6.2.1, 9.7.6.2.2, 9.7.6.2.6	H9
33 (c) (ii)	3	9.7.6.2.3, 9.7.6.2.4	H7, H10
33 (d) (i)	3	9.7.2.2.2.	H8
33 (d) (ii)	1	9.7.2.2.3	H8
33 (e)	6	9.7.5.2.1/2, 9.7.5.3.2, 9.7.4.2.2/3	H10
Question 34		From Quanta to Quarks	
34 (a) (i)	3	9.8.1.3.2, 9.8.1.2.5, 9.8.1.2.2	H7, H10
34 (a) (ii)	2	9.8.1.3.4	H7, H10
34 (b) (i)	3	9.8.3.2.11	H7
34 (b) (ii)	2	9.8.3.2.9, 9.8.3.2.4	H7
34 (c)	4	9.8.2.2.1, 9.8.4.2.3	H1, H8
34 (d) (i)	3	9.8.4.2.5	H9, H10
34 (d) (ii)	2	9.8.4.2.4	H2, H10
34 (e)	6	9.8.3.2.2, 9.8.3.2.6, 9.8.3.2.9, 9.8.1	H6
Question 35		The Age of Silicon	
35 (a) (i)	2	9.9.2.2.1	H9
35 (a) (ii)	2	9.9.3.2.1, 9.9.3.2.4, 9.9.2.2.6	H9
35 (a) (iii)	3	9.9.3.2.4, 9.9.2.3.3	H9
35 (a) (iv)	4	9.9.6.2.5, 9.9.6.2.7, 9.9.6.2.11, 9.9.6.2.8	H9
35 (a) (v)	3	9.9.6.3.2, 9.9.6	H9
35 (b)	2	9.9.4.2.1	H3, H9
35 (c)	3	9.9.5.3.1	H9
35 (d)	6	9.9.1.1	H3